

Question 1 (15 marks) Start a NEW page

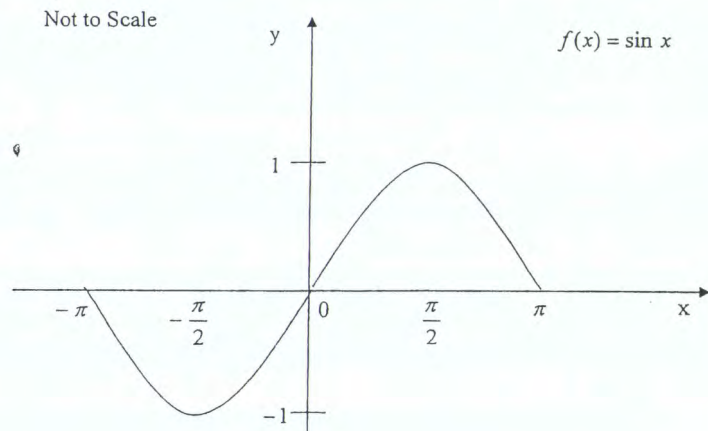
- (a) Let  $\alpha = 3 + 4i$  and  $\beta = 1 - i$   
Express in the form  $x + iy$  where  $x$  and  $y$  are real
- (i)  $\alpha\beta$  1
- (ii)  $\frac{\alpha}{\beta}$  2
- (iii)  $(\overline{\beta})^2$  2
- (b) Consider the equation  $z^2 + \gamma z + (2 - i) = 0$   
Find the complex number  $\gamma$ , given that  $2i$  is a root of the given equation. 2
- (c) Let  $\beta = -1 - i\sqrt{3}$
- (i) Express  $\beta$  in modulus-argument form. 2
- (ii) Express  $\beta^{-10}$  in modulus-argument form. 2
- (iii) Hence express  $\beta^{-10}$  in the form  $x + iy$  1
- (d) Shade the region in the number plane described by  
 $\frac{\pi}{3} < \arg z < \pi$  and  $1 \leq |z| \leq 3$  3

Question 2 (15 marks) Start a NEW page

- (a) (i) Find real numbers  $a$  and  $b$  such that  
$$\frac{1}{(3-x)(1+x)} \equiv \frac{a}{(3-x)} + \frac{b}{(1+x)}$$
 2
- (ii) Hence find  $\int \frac{1}{(3-x)(1+x)} dx$  2
- (b) Find  $\int \frac{1}{e^x + e^{-x}} dx$  3
- (c) By completing the square, find  
$$\int \frac{3}{\sqrt{x^2 + 6x + 13}} dx$$
 2
- (d) (i) Evaluate  $\int_0^{\frac{\pi}{2}} \frac{1}{1 + \cos x + \sin x} dx$ ,  
using the substitution  $t = \tan \frac{x}{2}$  3
- (ii) Hence evaluate  $\int_0^{\frac{\pi}{2}} \frac{x}{1 + \cos x + \sin x} dx$ ,  
using the substitution  $u = \frac{\pi}{2} - x$  3

**Question 3 (15 marks) Start a NEW page**

- (a) The diagram shows the graph of  $f(x) = \sin x$ , for  $-\pi \leq x \leq \pi$



- (b) (i) Sketch on the same number plane: 2

$$y = |x| - 2 \text{ and}$$

$$y = -x^2 + 3x$$

- (ii) Hence or otherwise solve  $\frac{|x| - 2}{-x^2 + 3x} > 0$  2

- (c)  $\alpha, \beta, \gamma$  are the roots of the equation  $x^3 + bx^2 + 12x + 4 = 0$ , where  $b$  is a real constant.

- (i) Find an equation with  $\alpha^{-1}, \beta^{-1}, \gamma^{-1}$  as roots. 2

- (ii) Hence, or otherwise, find the value of  $b$ , given that 2

$\alpha^{-1}, \beta^{-1}, \gamma^{-1}$  are in arithmetic progression.

Draw separate one-third page sketches of the graphs of the following:

(i)  $y = \frac{1}{f(x)}$  1

(ii)  $y = (f(x))^2$  2

(iii)  $|y| = f(x)$  2

(iv)  $y = \sin^{-1} f(x)$  2

Question 3 continues on page 4

End of Question 3

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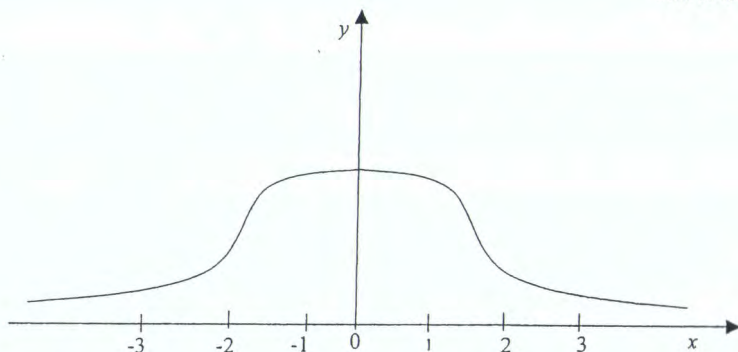
**Question 4 (15 marks) Start a NEW page**

- (a) The area bounded by the curve  $y = e^{-\frac{1}{2}x^2}$  and the  $x$ -axis, between  $x = 0$  and  $x = 2$  is rotated about the  $y$ -axis. Using the method of cylindrical shells, find the volume of the solid formed. (Leave your answer in exact form)

3

$$y = e^{-\frac{1}{2}x^2}$$

Not to Scale



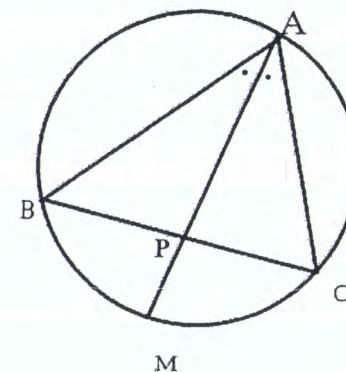
- (b) Consider the conic defined by the equation  $\frac{x^2}{19-l} + \frac{y^2}{7-l} = 1$

- (i) Determine the real value of  $l$  for which the equation defines: 2
- (α) an ellipse
- (β) an hyperbola
- (ii) Sketch the curve corresponding to  $l = 3$ . Show the foci and the  $x$  and  $y$  intercepts. You are NOT required to show the directrices. 2
- (iii) Describe how the shape of this curve changes as  $l$  increases from 3 towards 7. 1
- (iv) Describe the limiting shape of the curve as  $l$  approaches 7. 1

Question 4 continues on page 6

- (c) A circle is drawn to pass through the vertices of  $\triangle ABC$ .  $AM$  bisects  $\angle BAC$  and meets  $BC$  at  $P$  as shown in the diagram.

Not to Scale



Copy or trace this diagram onto your answer sheet.

- (i) Prove that  $\triangle ABM$  and  $\triangle ACP$  are similar. 2
- (ii) Prove that  $AB \cdot AC = AP \cdot AM$  1
- (iii) Hence prove that  $AB \cdot AC - BP \cdot PC = AP^2$  3

End of Question 4

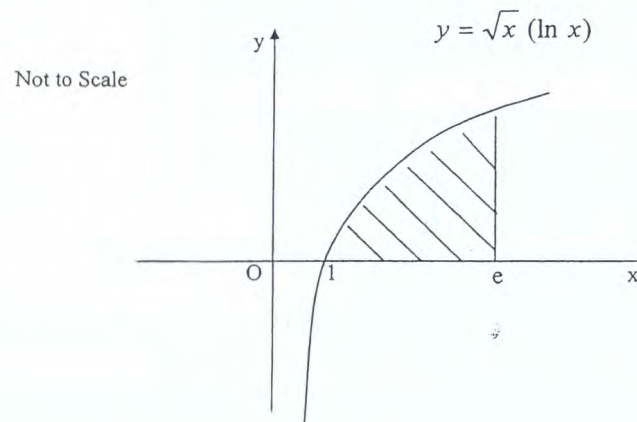


**Question 5 (15 marks) Start a NEW page**

- (a) (i) Show that  $2 \cos A \sin B = \sin(A + B) - \sin(A - B)$  1
- (ii) Hence, or otherwise, show that  $(1 + 2 \cos \theta + 2 \cos 2\theta + 2 \cos 3\theta) \sin \frac{\theta}{2} = \sin \frac{7\theta}{2}$  2
- (iii) Let  $A = 1 + 2 \cos \theta + 2 \cos 2\theta + 2 \cos 3\theta$  1  
 Show that if  $\theta = \frac{2\pi}{7}$ , then  $A = 0$
- (iv) Express  $A$  in terms of  $\cos \theta$  1  
 You may assume  $\cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta$ . Do NOT prove this.
- (v) Hence or otherwise prove that  $\cos \frac{2\pi}{7}$  1  
 is a solution of the polynomial equation  $8x^3 + 4x^2 - 4x - 1 = 0$

Question 5 continues on page 8

- (b) (i) Let  $I_n = \int_1^e x(\ln x)^n dx$ ,  $n = 0, 1, 2, \dots$  2  
 Use integration by parts to show that  $I_n = \frac{e^2}{2} - \frac{n}{2} I_{n-1}$ ,  $n = 1, 2, 3, \dots$
- (ii) The area bounded by the curve  $y = \sqrt{x} (\ln x)$ ,  $x \geq 1$ , the  $x$ -axis and the line  $x = e$  is rotated about the  $x$ -axis. Find the exact volume of the solid of revolution so formed. 3



- (c) 20 teachers volunteer to be considered for a special 50<sup>th</sup> anniversary celebrations committee. These consist of 4 Maths teachers, 4 Art teachers, 6 English teachers and 6 History teachers. 9 teachers are to be randomly chosen to form the committee.
- (i) If there are no restrictions, in how many ways can the 9 teachers be chosen? 1
- (ii) In how many ways can the 9 teachers be chosen if no Art or Maths teachers are included? 1
- (iii) What is the probability that no more than 1 Maths teacher and no more than 1 Art teacher are on the committee? 2

End of Question 5

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**Question 6 (15 marks) Start a NEW page**

- (a) A body of mass  $m$  is falling from rest and experiences air resistance of  $kv^2$  per unit mass.  $k$  is a constant,  $g$  is gravity under acceleration and  $v$  is the velocity of the body.

(i) Show that the equation of motion of the body is given by 1  

$$\ddot{x} = g - kv^2$$

(ii) If  $V$  is the terminal velocity, show that  $V = \sqrt{\frac{g}{k}}$  1

(iii) Show that  $x = \frac{1}{2k} \ln \left( \frac{g}{g - kv^2} \right)$ , where  $x$  is the distance travelled. 4

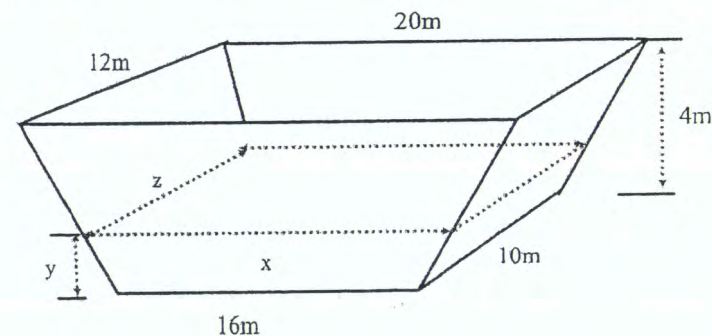
(iv) If  $W$  is the velocity of the body when it reaches the ground, show that the distance,  $S$ , fallen is given by 1

$$S = \frac{1}{2k} \ln \left( \frac{V^2}{V^2 - W^2} \right)$$

Question 6 continues on page 10

- (b) The Bundeena Voluntary Firefighters have a massive water storage tank. The tank has a rectangular base with sides 16 metres and 10 metres. Its top is also rectangular with dimensions 20 metres and 12 metres. The tank has a depth of 4 metres and each of its four side faces is an isosceles trapezium. Each horizontal cross section parallel to the base of the tank is a rectangle.

Not to Scale



- (i) Consider a cross section of the tank  $x$  metres by  $z$  metres and height  $y$  above the base. Show that the area of this cross section is given by 3

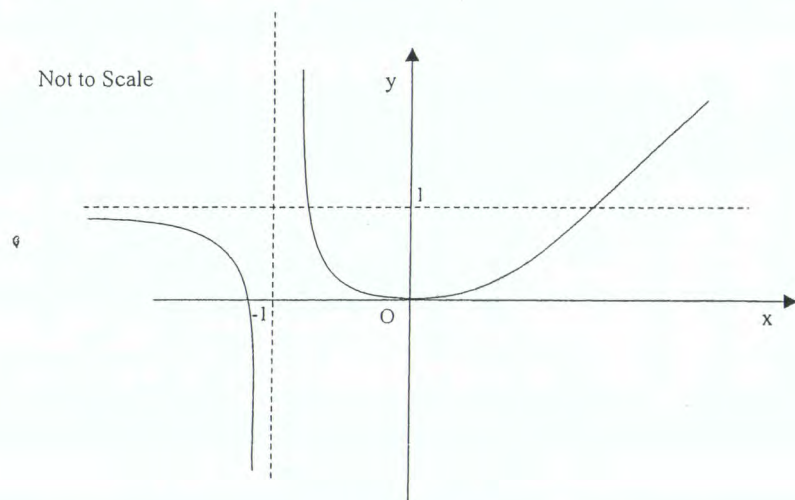
$$\frac{1}{2}y^2 + 18y + 160$$

- (ii) Hence find the volume of the tank. 2

Question 6 continues on page 11

(c)

3



The diagram shows the graph of  $y = f'(x)$ , the gradient function of  $y = f(x)$

Copy or trace this diagram onto your answer sheet.

Sketch, on a separate number plane and underneath your copy, the graph of  $y = f(x)$ , given that  $f(0) = 1$  and  $f(x) < 0$  for  $x < -1$ . Show all relevant asymptotes.

End of Question 6

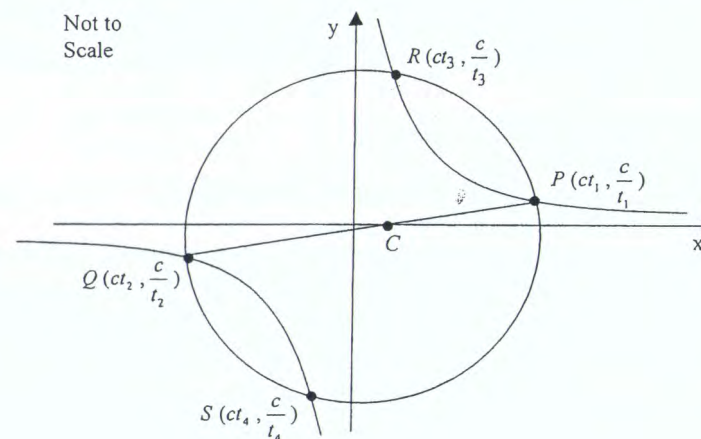
### Question 7 (15 marks) Start a NEW page

- (a) Let  $P$ ,  $Q$ ,  $R$  and  $S$  be four points with parameters  $t_1$ ,  $t_2$ ,  $t_3$  and  $t_4$  on the hyperbola  $x = ct$ ,  $y = \frac{c}{t}$ , as shown in the diagram.

The hyperbola is intersected by a circle whose equation is

$$(x - g)^2 + y^2 = r^2$$

$P$ ,  $Q$ ,  $R$  and  $S$  lie on the circle, as shown and  $C$  is the centre of the circle.



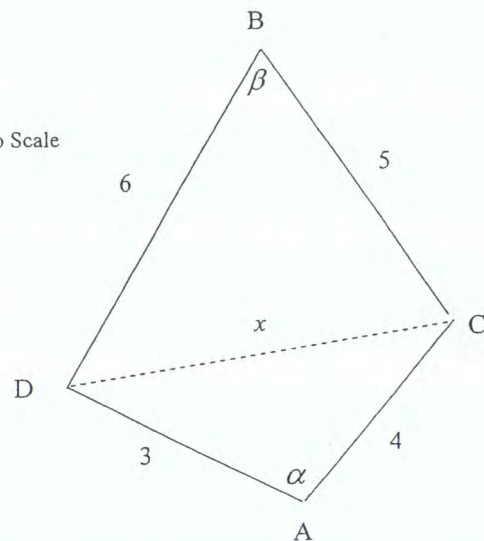
- (i) Show that  $t_1 + t_2 + t_3 + t_4 = \frac{2g}{c}$  2
- (ii) If  $C$  is the midpoint of  $PQ$ , show that  $t_1 + t_2 = \frac{2g}{c}$  2
- (iii) Hence, or otherwise, show that the origin is the midpoint of  $RS$ . 2

Question 7 continues on page 13



(b)

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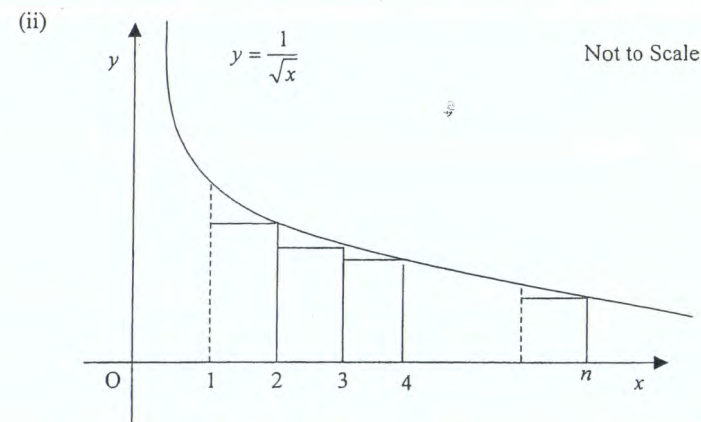
The quadrilateral ACBD has sides  $AC = 4$ ,  $CB = 5$ ,  $BD = 6$  and  $DA = 3$ .  
 $\angle DAC = \alpha$  and  $\angle DBC = \beta$ .  $DC = x$ , as shown.

- |       |                                                                                                                                                              |   |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| (i)   | Show that the area of ACBD is given by<br>$A = 6 \sin \alpha + 15 \sin \beta$                                                                                | 1 |
| (ii)  | By equating expressions for $x^2$ , show that<br>$2 \cos \alpha - 5 \cos \beta = -3$                                                                         | 2 |
| (iii) | Differentiate the identity in part (ii) implicitly,<br>with respect to $\alpha$ ,<br>to find an expression for $\frac{d\beta}{d\alpha}$ .                    | 1 |
| (iv)  | Prove that the area of the quadrilateral is a maximum when<br>the quadrilateral is cyclic. You need NOT prove the<br>relevant stationary point is a maximum. | 3 |
| (v)   | If the quadrilateral is cyclic, find the exact area.                                                                                                         | 2 |

End of Question 7

## Question 8 (15 marks) Start a NEW page

- |     |      |                                                                                                                                                                                                                   |   |
|-----|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| (a) | (i)  | Find a general solution for the equation<br>$\cos A = \cos B$<br>Write the answer in terms of $A$ and $B$ .                                                                                                       | 1 |
|     | (ii) | Hence, or otherwise, solve<br>$\cos 2\theta = \sin 3\theta$ , for $0 \leq \theta \leq \pi$                                                                                                                        | 3 |
| (b) | (i)  | Use Mathematical Induction to show that<br>$\sum_{r=1}^n \frac{1}{\sqrt{r}} > 2(\sqrt{n+1} - 1)$ , where $n = 1, 2, 3, \dots$<br>You may assume the inequality:<br>$2k+3 > 2\sqrt{(k+1)(k+2)}$ Do NOT prove this. | 3 |



Use the graph of  $y = \frac{1}{\sqrt{x}}$  to show that

$$\sum_{r=1}^n \frac{1}{\sqrt{r}} < 1 + \int_1^n \frac{1}{\sqrt{x}} dx$$

- |       |                                                     |   |
|-------|-----------------------------------------------------|---|
| (iii) | Hence show that                                     | 3 |
|       | $198 < \sum_{r=1}^{10000} \frac{1}{\sqrt{r}} < 199$ |   |

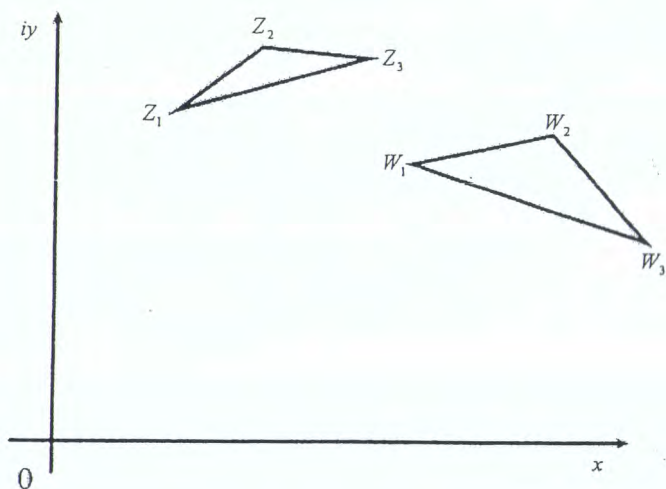
Question 8 continues on page 15

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- (c) The points  $Z_1, Z_2, Z_3$  represent the complex numbers  $z_1, z_2, z_3$  and the points  $W_1, W_2, W_3$  represent the complex numbers  $w_1, w_2, w_3$ .

If  $\frac{z_2 - z_1}{z_3 - z_1} = \frac{w_2 - w_1}{w_3 - w_1}$  prove that  $\triangle Z_1 Z_2 Z_3$  and  $\triangle W_1 W_2 W_3$  are similar.

Not to Scale



End of Paper

①  $z = 3 + 4i$ ,  $\beta = 1 - i$

(i)  $\alpha\beta = (3 + 4i)(1 - i)$   
 $= 3 - 3i + 4i + 4$   
 $= 7 + i$

(ii)  $\frac{\alpha}{\beta} = \frac{3 + 4i}{1 - i} \times \frac{1 + i}{1 + i}$   
 $= \frac{3 + 3i + 4i - 4}{1 + 1}$   
 $= \frac{-1 + 7i}{2}$   
 $= -\frac{1}{2} + \frac{7}{2}i$

(iii)  $(\frac{\alpha}{\beta})^2 = (1 + i)^2$   
 $= 1 + 2i - 1$   
 $= 2i$

Solutions.

①

②

②

(b)  $z + v + w + (2 - i) = 0$

Sub  $z = 2i$

$(2i) + 8 + 2i + 2 - i = 0$

$-4 + 2i + 2 - i = 0$

$-2 + 2i - i = 0$

$\times$  by  $i$

$-2i - 2 + 1 = 0$

$+2i = 1 - 2i$

$2(x + iy) = 1 - 2i$

$2x = 1$ ,  $2y = -2$

$x = \frac{1}{2}$ ,  $y = -1$

$z = \frac{1}{2} - i$

(c)  $\beta = -1 - i\sqrt{3}$

(i)  $|\beta| = \sqrt{1 + 3}$

$= 2$

$\tan \theta = \frac{1}{\sqrt{3}}$

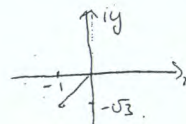
$\Rightarrow \theta = \frac{\pi}{3}$

$\arg \beta = -\frac{2\pi}{3}$

$\beta = 2 \operatorname{cis}(-\frac{2\pi}{3})$

(ii)  $\beta^{-10} = 2^{-10} \operatorname{cis}(\frac{20\pi}{3})$   
 $= 2^{-10} \operatorname{cis}(\frac{2\pi}{3})$

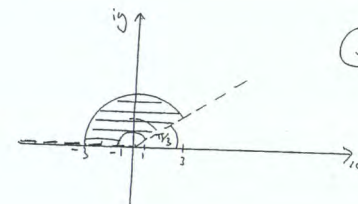
(iii)  $\beta^{-10} = 2^{-10} (\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3})$   
 $= 2^{-10} (\frac{-1}{2} + i \frac{\sqrt{3}}{2})$   
 $= -2^{-11} + i(\sqrt{3})2^{-11}$



②

① (d)

$\arg z > \frac{\pi}{3}$ ,  $1 \in \operatorname{Re} z \leq 3$



③

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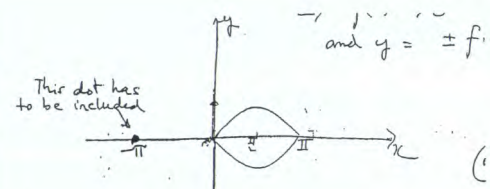
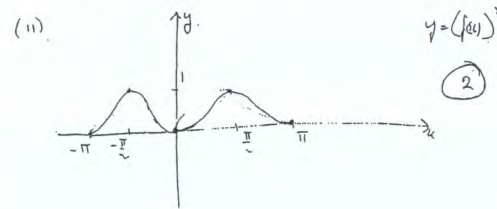
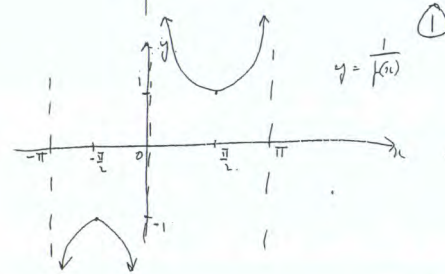
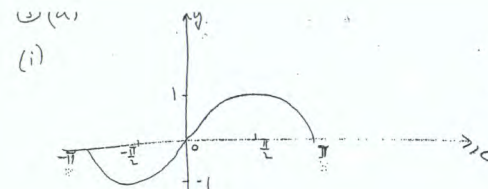
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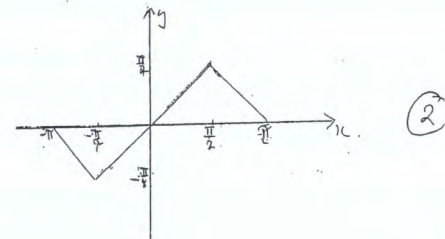
(a) (i)  $\frac{1}{(3-x)(1+x)} = \frac{A}{3-x} + \frac{B}{1+x}$   
 $= \frac{A(1+x) + B(3-x)}{(3-x)(1+x)}$   
 $= \frac{A(1+x) + B(3-x)}{(3-x)(1+x)}$   
 $1 = A(1+x) + B(3-x)$   
 $\text{Let } x = -1: 1 = 4B \Rightarrow B = \frac{1}{4}$   
 $\text{Let } x = 3: 1 = 4A \Rightarrow A = \frac{1}{4}$

(ii)  $\int \frac{1}{(3-x)(1+x)} dx = \frac{1}{4} \int \frac{1}{3-x} + \frac{1}{1+x} dx$   
 $= \frac{1}{4} \ln|3-x| + \frac{1}{4} \ln|1+x| + C$   
 $= \frac{1}{4} \ln \left| \frac{1+x}{3-x} \right| + C$

(b)  $\int \frac{e^x + e^{-x}}{e^x + 1} dx$   
 $= \int \frac{1}{e^x + e^{-x}} dx$   
 $= \int \frac{e^x}{e^{2x} + 1} dx$   
 $\text{Let } u = e^x$   
 $\frac{du}{dx} = e^x$   
 $du = e^x dx$   
 $= \int \frac{du}{u^2 + 1}$   
 $= \tan^{-1} u + C$   
 $= \tan^{-1} e^x + C$



(iv)  $y = \sin^{-1}(\sin x)$



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(2)  $\int \frac{3}{\sqrt{x^2 + 6x + 13}} dx$   
 $= \int \frac{3}{\sqrt{x^2 + 6x + 9 - 4 + 13}} dx$   
 $= \int \frac{3}{\sqrt{(x+3)^2 + 4}} dx$

$x^2 + 6x + 13$

$= 3 \ln \left| x+3 + \sqrt{(x+3)^2 + 4} \right| + C$

(d) (i)  $\int_0^{\pi/2} \frac{1}{1 + \cos x + \sin x} dx = \int_0^1 \frac{1}{1 + \frac{1-t^2}{1+t^2} + \frac{2t}{1+t^2}} \cdot \frac{2dt}{1+t^2}$   
 $\left[ \text{where } t = \tan \frac{x}{2} \right]$   
 $\frac{dx}{dt} = \frac{2}{1+t^2}$

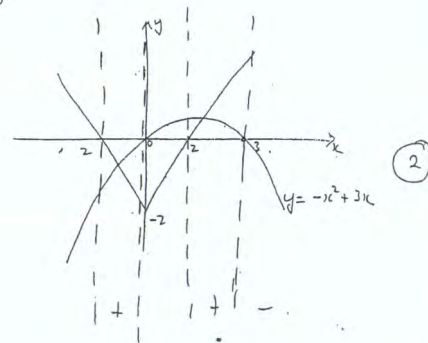
$= \int_0^1 \frac{1+t^2}{1+t^2+1-t^2+2t} \times \frac{2dt}{1+t^2}$   
 $= \int_0^1 \frac{1+t^2}{2(1+t)} \times \frac{2dt}{1+t^2}$   
 $= \int_0^1 \frac{dt}{1+t}$   
 $= \left[ \ln|1+t| \right]_0^1$   
 $= \ln 2$

(ii)  $I = \int_0^{\pi/2} \frac{x}{1 + \cos x + \sin x} dx$   
 $\text{Let } u = \frac{\pi}{2} - x$   
 $\frac{du}{dx} = -1$

$I = \int_{\pi/2}^0 \frac{\frac{\pi}{2} - u}{1 + \cos(\frac{\pi}{2} - u) + \sin(\frac{\pi}{2} - u)} (-1) du$   
 $= \int_0^{\pi/2} \frac{\frac{\pi}{2} - u}{1 + \sin u + \cos u} du$   
 $= \int_0^{\pi/2} \frac{\frac{\pi}{2} du}{1 + \sin u + \cos u} - \int_0^{\pi/2} \frac{u du}{1 + \sin u + \cos u}$   
 $= \frac{\pi}{2} \int_0^{\pi/2} \frac{du}{1 + \sin u + \cos u}$   
 $= \frac{\pi}{2} \times \ln 2$ , from (i)

$I = \frac{\pi}{2} \ln 2$

b) (i)

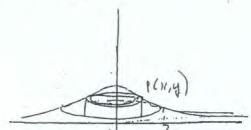


(ii)  $-2 < x < 0, 2 < x < 3$

(c)  $x^3 + 6x^2 + 12x + 4 = 0$  ... (1)  
 (i) Equat with roots  $\frac{1}{x}, \frac{1}{x}, \frac{1}{x}$  will be:  
 $\left(\frac{1}{x}\right)^3 + 6\left(\frac{1}{x}\right)^2 + 12\left(\frac{1}{x}\right) + 4 = 0$   
 $\times \text{ by } x^3$   
 $1 + 6x + 12x^2 + 4x^3 = 0$   
 $\therefore 4x^3 + 12x^2 + 6x + 1 = 0$

(ii) If  $\frac{1}{x}, \frac{1}{x}, \frac{1}{x}$  are in A.P.  
 $\text{Let } \frac{1}{x} = \frac{1}{y} - d, \frac{1}{x} = \frac{1}{y} + d$   
 $\therefore \sum \frac{1}{x} = -\frac{3}{4} = -\frac{12}{4} = -3$   
 $\therefore \beta = -1$   
 $\text{Sub. for } x = -1 \text{ in (1)}$

$\therefore -1 + 6 - 12 + 4 = 0$   
 $\therefore b = 9$



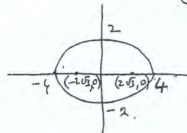
$$\begin{aligned}
 A &= \pi (R^2 - r^2) \\
 &= \pi (R^2 - r^2) \\
 &= \pi \int_{-r}^r \int_{-\sqrt{R^2-x^2}}^{\sqrt{R^2-x^2}} dy dx \\
 &= 2\pi \int_{-r}^r \sqrt{R^2-x^2} dx \\
 &= 2\pi \int_{-r}^r \sqrt{R^2-x^2} dx \\
 &= 2\pi \left[ \frac{x}{2} \sqrt{R^2-x^2} + \frac{R^2}{2} \sin^{-1} \frac{x}{R} \right]_{-r}^r \\
 &= 2\pi \left[ \frac{r}{2} \sqrt{R^2-r^2} + \frac{R^2}{2} \sin^{-1} \frac{r}{R} \right] \\
 &= 2\pi \left( 1 - e^{-1} \right) \text{ units}^2
 \end{aligned}$$

let  $u = x^2$   
 $\frac{du}{dx} = 2x$   
 $\therefore x dx = \frac{du}{2}$

(3)

$$\begin{aligned}
 19 > l & \quad l < 7 \\
 \text{and } 7 - l > 0 & \\
 7 > l & \\
 (p) \quad 19 - l > 0 & \quad (2) \\
 19 > l & \\
 \text{and } 7 - l < 0 & \\
 l > 7 & \\
 7 < l < 19 &
 \end{aligned}$$

$$\begin{aligned}
 (ii) \quad \frac{x^2}{16} + \frac{y^2}{4} &= 0 \\
 b^2 &= a^2 (1 - e^2) \\
 4 &= 16 (1 - e^2) \\
 \frac{1}{4} &= 1 - e^2 \\
 e^2 &= \frac{3}{4} \\
 e &= \frac{\sqrt{3}}{2} \\
 \text{foci } x &= \pm 2\sqrt{3}
 \end{aligned}$$



(2)

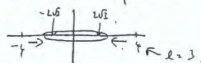
$$\begin{aligned}
 (i) \quad \sin(A+B) - \sin(A-B) &= \sin A \cos B + \cos A \sin B - \sin A \cos B + \cos A \sin B \\
 &= 2 \cos A \sin B \\
 (ii) \quad (1 + 2 \cos \theta + 2 \cos 2\theta + 2 \cos 3\theta) \sin \frac{\theta}{2} &= \sin \frac{\theta}{2} + 2 \cos \theta \sin \frac{\theta}{2} + 2 \cos 2\theta \sin \frac{\theta}{2} + 2 \cos 3\theta \sin \frac{\theta}{2} \\
 &= \sin \frac{\theta}{2} + \sin \frac{3\theta}{2} - \sin \frac{\theta}{2} + \sin \frac{5\theta}{2} - \sin \frac{3\theta}{2} \\
 &= \sin \frac{5\theta}{2}
 \end{aligned}$$

$$\begin{aligned}
 (iii) \quad \text{From (ii)} \quad A \sin \frac{\theta}{2} &= \sin \frac{5\theta}{2} \\
 \text{if } \theta &= \frac{2\pi}{5} \\
 A \sin \frac{2\pi}{5} &= \sin \frac{2\pi}{5} \\
 A \sin \frac{\pi}{5} &= \sin \frac{\pi}{5} = 0 \\
 \Rightarrow A &= 0 \quad \text{since } \sin \frac{\pi}{5} \neq 0
 \end{aligned}$$

$$\begin{aligned}
 (iv) \quad A &= 1 + 2 \cos \theta + 2 \cos 2\theta + 2 \cos 3\theta \\
 &= 1 + 2 \cos \theta + 2 (2 \cos^2 \theta - 1) \\
 &\quad + 2 (4 \cos^3 \theta - 3 \cos \theta) \\
 &= 1 + 2 \cos \theta + 4 \cos^2 \theta - 2 \\
 &\quad + 8 \cos^3 \theta - 6 \cos \theta \\
 &= 8 \cos^3 \theta + 4 \cos^2 \theta - 4 \cos \theta - 1 \\
 (v) \quad \text{If } \theta &= \frac{4\pi}{5}, A = 0 \\
 \text{let } x &= \cos \theta \text{ in (i)} \\
 \therefore A &= 8x^3 + 4x^2 - 4x - 1 = 0 \quad \text{when } \theta = \frac{4\pi}{5} \\
 \text{is } \cos \frac{4\pi}{5} & \text{ is a root of the given cubic}
 \end{aligned}$$

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$$\begin{aligned}
 (iii) \quad A & \text{ increases from 3 to 7} \\
 a^2 &\rightarrow 19 - 7 = 12 \\
 b^2 &\rightarrow 7 - 7 = 0 \\
 \text{as } a &\rightarrow 2\sqrt{3}, b \rightarrow 0 \\
 \text{so the curve becomes flatter and narrower.}
 \end{aligned}$$

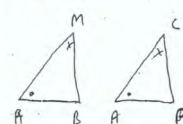


$$\begin{aligned}
 (iv) \quad A & \text{ straight line on the } x \text{ axis} \\
 \text{between } x &= -2\sqrt{3} \text{ and } x = +2\sqrt{3}, \\
 \text{ie between the original foci.}
 \end{aligned}$$

(1)

$$\begin{aligned}
 (i) \quad \text{In } \Delta s \quad ABM &+ ABC \\
 \angle AMB &= \angle ACB \text{ (given)} \\
 \angle BAM &= \angle BCA \text{ (Lr subtends at the circumference by the same arc)} \\
 \therefore \Delta ABM &\sim \Delta ABC \text{ (equiangular)}
 \end{aligned}$$

$$\begin{aligned}
 (ii) \quad \frac{AB}{AP} &= \frac{AM}{AC} \text{ (corresponding sides in } \Delta s) \\
 \therefore AB, AC &= AP, AM \\
 (iii) \quad BP, PC &= AP, PM \text{ (intersecting chords in same ratio)}
 \end{aligned}$$



$$\begin{aligned}
 AB, AC &= BP, PC = AP, PM \text{ (intersecting chords in same ratio)} \\
 AB, AC &= BP, PC = AP, PM, \text{ from (ii)} \\
 &= AP (AM - PM) \\
 &= AP \times AP \\
 &= AP^2
 \end{aligned}$$

(3)

$$\begin{aligned}
 (b) \quad (i) \quad I_n &= \int_1^e x (\ln x)^n dx \\
 \text{let } u &= (\ln x)^n, v' = x \Rightarrow v = \frac{1}{2} x^2 \\
 I_n &= \left[ \frac{1}{2} x^2 (\ln x)^n \right]_1^e - \int_1^e \frac{1}{2} x^2 \cdot n (\ln x)^{n-1} \times \frac{1}{x} dx \\
 &= \frac{1}{2} e^2 - \frac{n}{2} \int_1^e x (\ln x)^{n-1} dx \\
 &= \frac{1}{2} e^2 - \frac{n}{2} I_{n-1}
 \end{aligned}$$

$$\begin{aligned}
 (ii) \quad V &= \pi \int_1^e x (\ln x)^n dx \\
 &= \pi I_n = \left( \frac{1}{2} e^2 - \frac{n}{2} I_{n-1} \right) \pi \\
 &\quad + \left( \frac{1}{2} e^2 - \frac{n}{2} I_{n-1} \right) \pi \\
 I_0 &= \int_1^e x dx \\
 &= \left[ \frac{1}{2} x^2 \right]_1^e = \frac{1}{2} (e^2 - 1) \\
 \therefore V &= \pi \left[ \frac{1}{2} e^2 - \frac{1}{2} e^2 + \frac{1}{2} \times \frac{1}{2} (e^2 - 1) \right] \\
 &= \frac{\pi}{4} [e^2 - 1] \text{ units}^3
 \end{aligned}$$

$$\begin{aligned}
 (c) \quad 20 &= 4M, 4A, 6E, 6H \\
 (i) \quad {}^{20}C_9 &= 167960 \\
 (ii) \quad {}^{12}C_9 &= 220
 \end{aligned}$$

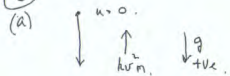
|        | no math | no A             | no math | no A | 12C9             | 12C8             | 12C7             | 12C6             |
|--------|---------|------------------|---------|------|------------------|------------------|------------------|------------------|
| 1 math | no A    | 4C1 x 12C8       | 1 math  | no A | 4C1 x 12C8       | 4C1 x 12C7       | 4C1 x 12C6       | 4C1 x 12C5       |
| 1 math | 1 A     | 4C1 x 4C1 x 12C6 | 1 math  | 1 A  | 4C1 x 4C1 x 12C5 | 4C1 x 4C1 x 12C4 | 4C1 x 4C1 x 12C3 | 4C1 x 4C1 x 12C2 |
|        |         |                  |         |      |                  |                  |                  |                  |

$$\begin{aligned}
 \text{Prob} &= \frac{12100}{167960} \\
 &= \frac{605}{8398}
 \end{aligned}$$

(2)



(6)



(i)  $m \frac{dv}{dt} = mg - kv$  ①

$kv = g$

(ii) Terminal vel approached when  $\frac{dv}{dt} = 0$  ①

$0 = g - kv$   
 $v = \frac{g}{k}$  (+ve  $\therefore$  particle going down)

(iii)  $v \frac{dv}{dx} = g - kv$   
 $\frac{dv}{dx} = \frac{g}{v} - kv = \frac{g - kv^2}{v}$  ④

$\frac{dx}{dv} = \frac{v}{g - kv^2}$   
 $x = \int \frac{v}{g - kv^2} dv$  let  $v^2 = u$   
 $\frac{dx}{du} = \frac{1}{2v}$

$= \frac{1}{2} \int \frac{du}{g - ku}$   
 $= \frac{1}{2k} \ln(g - ku) + C$   
 $= \frac{1}{2k} \ln(g - kv^2) + C$

when  $u=0, v=0$   
 $0 = -\frac{1}{2k} \ln g + C$   
 $C = \frac{1}{2k} \ln g$

$x = -\frac{2k}{g} \ln(g - kv^2) + C$

$= \frac{1}{2k} \ln\left(\frac{g}{g - kv^2}\right)$

(iv) when  $v = W$   
 $S = \frac{1}{2k} \ln\left(\frac{g}{g - kW^2}\right)$

From (ii),  $v = \sqrt{\frac{g}{k}}$   
 $\therefore g = kv^2$

$S = \frac{1}{2k} \ln\left(\frac{kv^2}{kv^2 - kW^2}\right)$   
 $= \frac{1}{2k} \ln\left(\frac{v^2}{v^2 - W^2}\right)$

①

$r_1, r_2, r_3, r_4$  are roots of the equation coming from sub  
 $x = ct + y = \frac{c}{t}$  in  
 $(x-g)^2 + y^2 = r^2$

$(ct-g)^2 + \left(\frac{c}{t}\right)^2 = r^2$  ②

$c^2 t^2 - 2gct + g^2 + \frac{c^2}{t^2} = r^2$

$c^2 t^4 - 2gct^3 + g^2 t^2 + c^2 = r^2 t^2$

$c^2 t^4 - 2gct^3 + g^2 t^2 - r^2 t^2 + c^2 = 0$

$t_1 + t_2 + t_3 + t_4 = \frac{2gc}{c^2} = \frac{2g}{c}$

(ii)  $C = (g, 0)$

$\therefore g = \frac{ct_1 + ct_2}{2}$

in  $t_1 + t_2 = \frac{2g}{c}$

②

(iii) Adjoint of  $H = M = \begin{bmatrix} \frac{c(t_1+t_2)}{2} & \frac{1}{2}\left(\frac{c}{t_1} + \frac{c}{t_2}\right) \end{bmatrix}$   
 $= (x_1, x_2)$  ②

$x_1 = \frac{c}{2} \left( \frac{2g}{c} - t_1 - t_2 \right)$   
 $= \frac{c}{2} \left( \frac{2g}{c} - \frac{2g}{c} \right) = 0$   
 $y_1 = \frac{c}{2} \left( \frac{t_1+t_2}{t_1 t_2} \right)$   
 $= 0$ , since  $t_1 t_2 = \frac{2g}{c} - (t_1 + t_2) = 0$  from (ii)

$15 \sin \alpha + 6 \sin \alpha$  ①

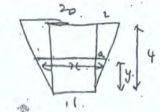
(ii)  $x^2 = 36 + 25 - 2 \times 70 \cos \beta = 9 + 16 - 2 \times 12 \cos \beta$   
 $24 \cos \alpha = 60 \cos \beta = -36$   
 $\therefore 2 \cos \alpha = 5 \cos \beta = -2$  ①

(iii)  $-2 \sin \alpha + 5 \sin \beta \cdot \frac{d\beta}{d\alpha} = 0$   
 $\frac{d\beta}{d\alpha} = \frac{2 \sin \alpha}{5 \sin \beta}$  ①

(iv)  $\frac{dA}{d\alpha} = 6 \cos \alpha + 15 \cos \beta \cdot \frac{d\beta}{d\alpha} = 0$  for a max  
 $\therefore 0 = 6 \cos \alpha + 15 \cos \beta \cdot \frac{2 \sin \alpha}{5 \sin \beta}$   
 $0 = \frac{6}{\sin \beta} (\sin \beta \cos \alpha + \sin \alpha \cos \beta)$  ③

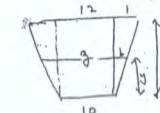
$0 = \sin(\alpha + \beta)$   
 $\Rightarrow \alpha + \beta = 0, \pi, 2\pi \rightarrow 0$  impossible since  $\alpha, \beta$  are angles  
 $\therefore \alpha + \beta = \pi$   
 $\therefore$  the quad is cyclic (opp  $\angle$ s supp)

b) (i)



$\frac{x}{y} = \frac{20}{4}$   
 $x = 5y$   
 $x - 2a = 16$   
 $a = \frac{x-16}{2} = \frac{5y-16}{2}$   
 $y = x - 16$   
 $16 = (y+16)$

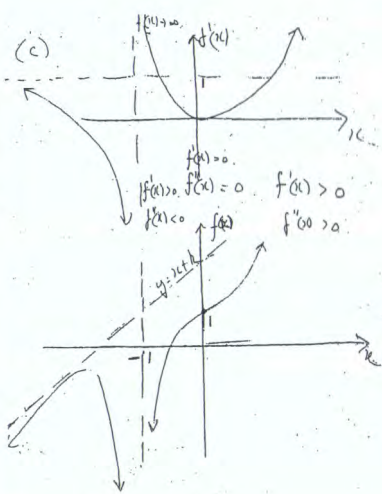
③



$\frac{1}{y} = \frac{1}{4}$   
 $4y = y$   
 $2y - 10 = y$   
 $y = \frac{10}{2} = 5$   
 $g - 2b = 10$   
 $2b = g - 10$

Area =  $(y+16)\left(\frac{y}{2} + 10\right)$   
 $= \frac{1}{2} y^2 + 17y + 160$  ②

(ii)  $Vol = \lim_{dy \rightarrow 0} \sum_{y=0}^y \frac{1}{2} y^2 + 17y + 160 \cdot dy$   
 $= \int_0^y \left( \frac{1}{2} y^2 + 17y + 160 \right) dy$   
 $= \left[ \frac{1}{6} y^3 + 9y^2 + 160y \right]_0^y$   
 $= \frac{1}{6} \times 64 + 9 \times 16 + 160 \times 4$   
 $= 794 \frac{2}{3}$  m<sup>3</sup>



p12

③

(v)  $2 \cos \alpha - 5 \cos \beta = -3$   
 $\therefore 2 \cos \alpha - 5 \cos(\pi - \alpha) = -3$   
 $2 \cos \alpha + 5 \cos \alpha = -3$   
 $7 \cos \alpha = -3$   
 $\cos \alpha = -\frac{3}{7}$  ✓

$A = 6 \sin \alpha + 15 \sin(\pi - \alpha)$   
 $= 6 \times \frac{\sqrt{40}}{7} + 15 \times \frac{\sqrt{40}}{7}$   
 $= \left( \frac{12}{7} + \frac{45}{7} \right) \sqrt{40}$   
 $= 6 \sqrt{40} \text{ units}^2$

②

⑧ (a) (i)  $\cos A = \cos B$   
 $\therefore A = 2n\pi \pm \cos^{-1}(\cos B)$  ①  
 $= 2n\pi \pm B, n \text{ is an integer}$

(ii)  $\cos 2\theta = \sin 3\theta, 0 \leq \theta \leq \pi$   
 $= \cos\left(\frac{\pi}{2} - 3\theta\right)$  ③  
 $\therefore 2\theta = 2n\pi \pm \left(\frac{\pi}{2} - 3\theta\right)$

$2\theta = 2n\pi + \frac{\pi}{2} - 3\theta$  or  $2\theta = 2n\pi - \frac{\pi}{2} + 3\theta$   
 $5\theta = 2n\pi + \frac{\pi}{2}$  or  $\theta = \frac{2n\pi + \frac{\pi}{2}}{5}$   
 $n=0 \Rightarrow \theta = \frac{\pi}{10}$   
 $n=1 \Rightarrow \theta = \frac{2\pi + \frac{\pi}{2}}{5} = \frac{5\pi + \frac{\pi}{2}}{10} = \frac{11\pi}{20}$   
 $n=2 \Rightarrow \theta = \frac{4\pi + \frac{\pi}{2}}{5} = \frac{9\pi}{10}$   
 $\theta = \frac{\pi}{10}, \frac{11\pi}{20}, \frac{9\pi}{10}$

Final solutions:  $\theta = \frac{\pi}{10}, \frac{11\pi}{20}, \frac{9\pi}{10}$

p14



(b) (i) Show  $\sum_{r=1}^n \frac{1}{\sqrt{r}} > 2(\sqrt{n+1} - 1)$

Let  $n=1$  : LHS = 1 RHS =  $2(\sqrt{2} - 1)$   
 $= 0.828$

True for  $n=1$

assume true for  $n=k$

i.e.  $\sum_{r=1}^k \frac{1}{\sqrt{r}} > 2(\sqrt{k+1} - 1)$  — (1)

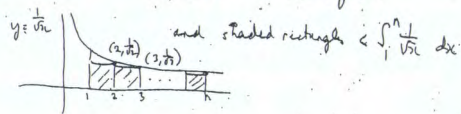
now true for  $n=k+1$

i.e.  $\sum_{r=1}^{k+1} \frac{1}{\sqrt{r}} > 2(\sqrt{k+2} - 1)$

proof:  $S_{k+1} = S_k + \frac{1}{\sqrt{k+1}}$   
 $> 2(\sqrt{k+1} - 1) + \frac{1}{\sqrt{k+1}}$  from (1)  
 $= \frac{2(k+1) - 2\sqrt{k+1} + 1}{\sqrt{k+1}}$   
 $= \frac{2k+3 - 2\sqrt{k+1}}{\sqrt{k+1}}$   
 $> \frac{2\sqrt{k+1}(k+1) - 2\sqrt{k+1}}{\sqrt{k+1}}$ , given property  
 $= 2(\sqrt{k+2} - 1)$

∴ true for  $n=1, 2, \dots, k, k+1$ , ∴ True  $\forall$  integers  $n$  by induction.

(ii)  $\sum_{r=1}^n \frac{1}{\sqrt{r}} < 1 + \frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \dots + \frac{1}{\sqrt{n}}$   
 $= 1 + \text{shaded rectangles}$

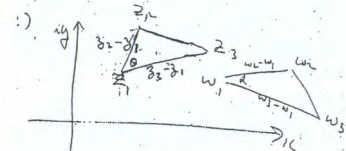


$\sum_{r=1}^n \frac{1}{\sqrt{r}} < 1 + \int_1^n \frac{1}{\sqrt{x}} dx$

(iii)  $\sum_{r=1}^{10000} \frac{1}{\sqrt{r}} < 1 + \int_1^{10000} \frac{1}{\sqrt{x}} dx$   
 $= 1 + [2\sqrt{x}]_1^{10000}$   
 $= 1 + 2 \times 100 - 2$   
 $= 199$

also  $\sum_{r=1}^{10000} \frac{1}{\sqrt{r}} > 2(\sqrt{10000+1} - 1)$  from (i)  
 $= 198.00\dots$

$\therefore 198 < \sum_{r=1}^{10000} \frac{1}{\sqrt{r}} < 199$



Let  $\frac{z_2 - z_1}{z_3 - z_1} = \frac{w_2 - w_1}{w_3 - w_1}$ , let  $\theta + 2\pi$  be  $\angle$  shown

then  $\left| \frac{z_2 - z_1}{z_3 - z_1} \right| = \left| \frac{w_2 - w_1}{w_3 - w_1} \right|$

$\frac{|z_2 - z_1|}{|z_3 - z_1|} = \frac{|w_2 - w_1|}{|w_3 - w_1|}$ , i.e. corresponding sides in same ratio

also  $\text{Arg} \left( \frac{z_2 - z_1}{z_3 - z_1} \right) = \text{Arg} \left( \frac{w_2 - w_1}{w_3 - w_1} \right)$ , since  $\frac{z_2 - z_1}{z_3 - z_1} = \frac{w_2 - w_1}{w_3 - w_1}$

i.e.  $\text{Arg}(z_2 - z_1) - \text{Arg}(z_3 - z_1)$   
 $= \text{Arg}(w_2 - w_1) - \text{Arg}(w_3 - w_1)$

i.e.  $\theta = \angle$

$\Delta z_1 z_2 z_3 \sim \Delta w_1 w_2 w_3$

(SAS similarity)